

ESP32 Smart Motion Lighting System

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Projektbeschreibung

Dieses Projekt demonstriert ein intelligentes Beleuchtungssystem mit ESP32, LDR-Lichtsensord, PIR-Bewegungssensord und PWM-Technologie. Wenn genug Umgebungslicht vorhanden ist, bleiben die LEDs ausgeschaltet. Wenn es dunkel ist und eine Bewegung erkannt wird, schalten die rote und die grüne LED auf maximale Helligkeit. Nach der Bewegung bleiben die LEDs für 10 Sekunden hell und wechseln danach in einen gedimmden Energiesparmodus.

Komponente	Verbindung / Funktion
ESP32 DevKit V1	Zentrale Steuerung
LDR Sensor AO	GPIO34 - Messung der Helligkeit
PIR Sensor OUT	GPIO27 - Bewegungserkennung
Rote + grüne LED	GPIO14 - gemeinsame PWM-Steuerung
Interne blaue LED	GPIO2 - Statusanzeige
PWM	Dynamische Helligkeitsregelung

Systemlogik

- Helle Umgebung - LEDs AUS.
- Dunkelheit ohne Bewegung - LEDs gedimmt.
- Dunkelheit mit Bewegung - LEDs maximale Helligkeit.
- Nach Bewegung - LEDs bleiben 10 Sekunden hell.
- Danach - Rückkehr zum gedimmden Energiesparmodus.

MicroPython Code

```
from machine import Pin, ADC, PWM
import time

# LEDs connected together on GPIO14
leds = PWM(Pin(14))
leds.freq(1000)

# Internal ESP32 blue LED
internal_led = Pin(2, Pin.OUT)

# LDR light sensor on GPIO34
light_sensor = ADC(Pin(34))
light_sensor.atten(ADC.ATTN_11DB)
light_sensor.width(ADC.WIDTH_12BIT)

# PIR motion sensor on GPIO27
motion_sensor = Pin(27, Pin.IN)

# Keep LEDs bright for 10 seconds after motion
motion_timeout = 10
last_motion_time = 0

print("Smart Motion Lighting System Running...")
```

```

while True:
    light_value = light_sensor.read()
    motion = motion_sensor.value()

    print("Light:", light_value)
    print("Motion:", motion)

    # Bright environment
    if light_value < 1800:
        leds.duty(0)
        internal_led.off()
        print("Bright Environment -> LEDs OFF")

    # Dark environment
    else:
        if motion == 1:
            last_motion_time = time.time()
            leds.duty(1023)
            internal_led.on()
            print("Motion Detected -> LEDs HIGH")

        else:
            elapsed = time.time() - last_motion_time

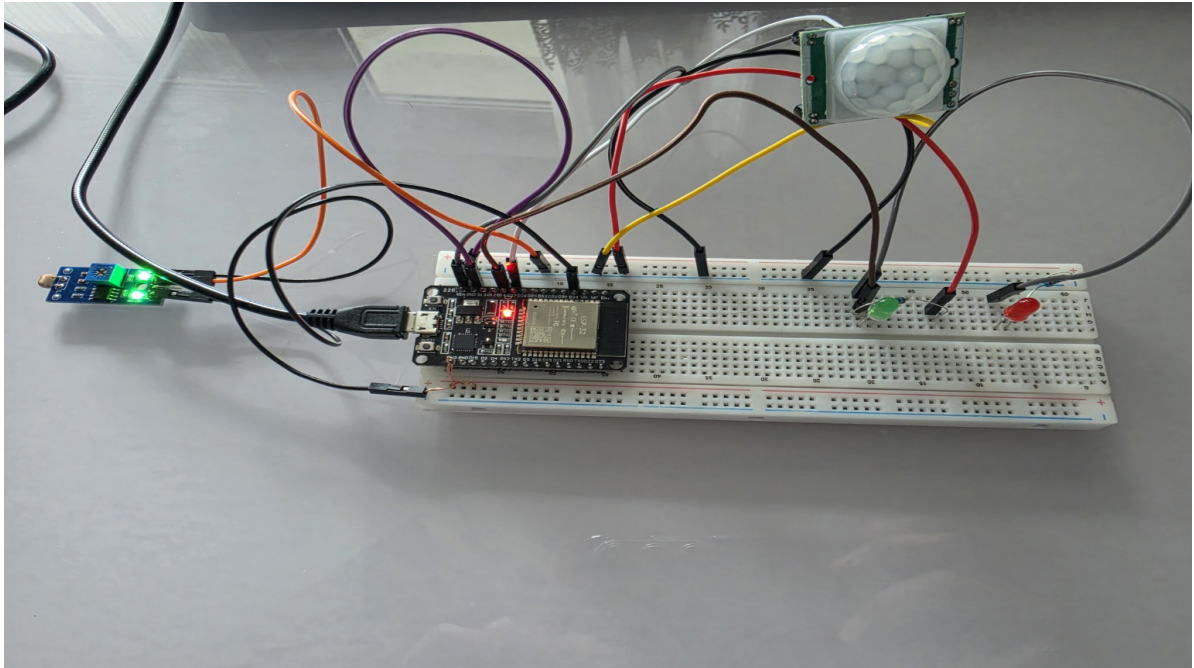
            if elapsed < motion_timeout:
                leds.duty(1023)
                internal_led.on()
                print("Waiting Timeout -> LEDs STILL ON")
            else:
                leds.duty(250)
                internal_led.on()
                print("No Motion -> LEDs DIM")

    print("-----")
    time.sleep(0.5)

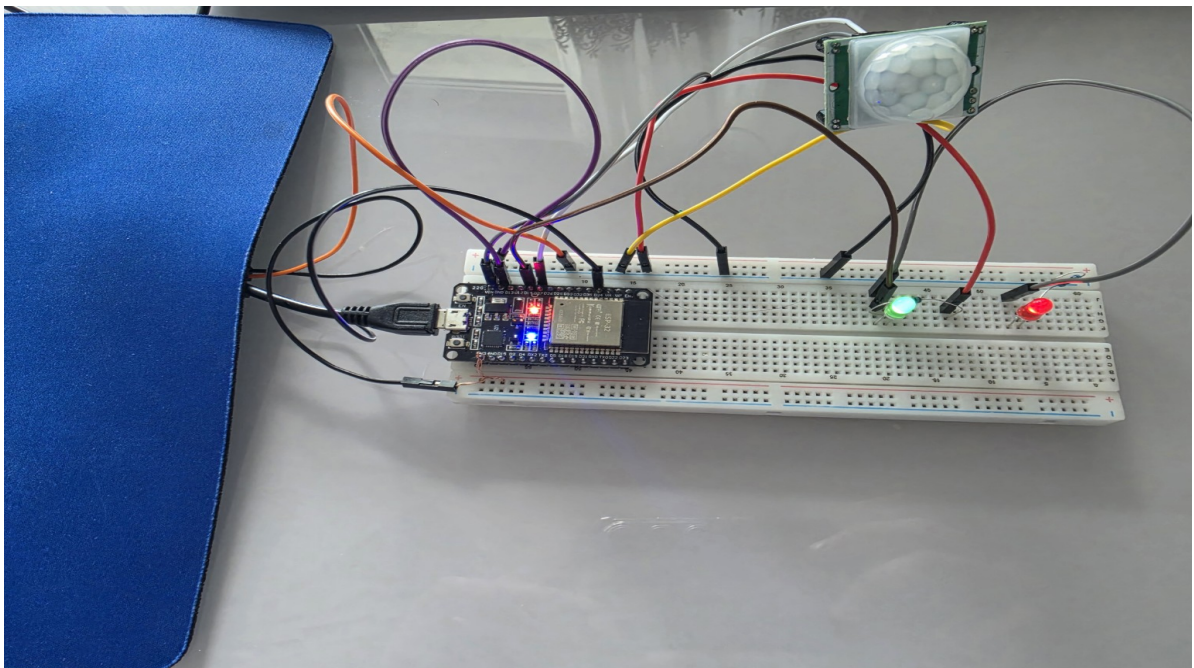
```

Projektbilder

Helle Umgebung - LEDs AUS



Dunkelheit + Bewegung - LEDs EIN / maximale Helligkeit



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Project Description

This project demonstrates an intelligent lighting system using ESP32, an LDR light sensor, a PIR motion sensor, and PWM technology. When enough ambient light is available, the LEDs remain OFF. When the environment is dark and motion is detected, the red and green LEDs switch to maximum brightness. After motion is detected, the LEDs stay bright for 10 seconds and then return to a dim energy-saving mode.

Component	Connection / Function
ESP32 DevKit V1	Main controller
LDR Sensor AO	GPIO34 - light intensity measurement
PIR Sensor OUT	GPIO27 - motion detection
Red + green LED	GPIO14 - shared PWM control
Internal blue LED	GPIO2 - status indicator
PWM	Dynamic brightness control

System Logic

- Bright environment - LEDs OFF.
- Dark environment without motion - LEDs dimmed.
- Dark environment with motion - LEDs at maximum brightness.
- After motion - LEDs stay bright for 10 seconds.
- After timeout - return to dim energy-saving mode.

Practical Applications

This project can be used as a model for smart home lighting, smart corridor lighting, security lighting, garage lighting, solar-powered lighting systems, renewable energy applications, and energy-efficient automation systems. It is especially useful for reducing energy consumption by activating strong lighting only when it is needed.

MicroPython Code

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import time

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leds.freq(1000)

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motion_sensor = Pin(27, Pin.IN)

# Keep LEDs bright for 10 seconds after motion
motion_timeout = 10
last_motion_time = 0

print("Smart Motion Lighting System Running...")

while True:
    light_value = light_sensor.read()
    motion = motion_sensor.value()

    print("Light:", light_value)
    print("Motion:", motion)

    # Bright environment
    if light_value < 1800:
        leds.duty(0)
        internal_led.off()
        print("Bright Environment -> LEDs OFF")

    # Dark environment
    else:
        if motion == 1:
            last_motion_time = time.time()
            leds.duty(1023)
            internal_led.on()
            print("Motion Detected -> LEDs HIGH")

        else:
            elapsed = time.time() - last_motion_time

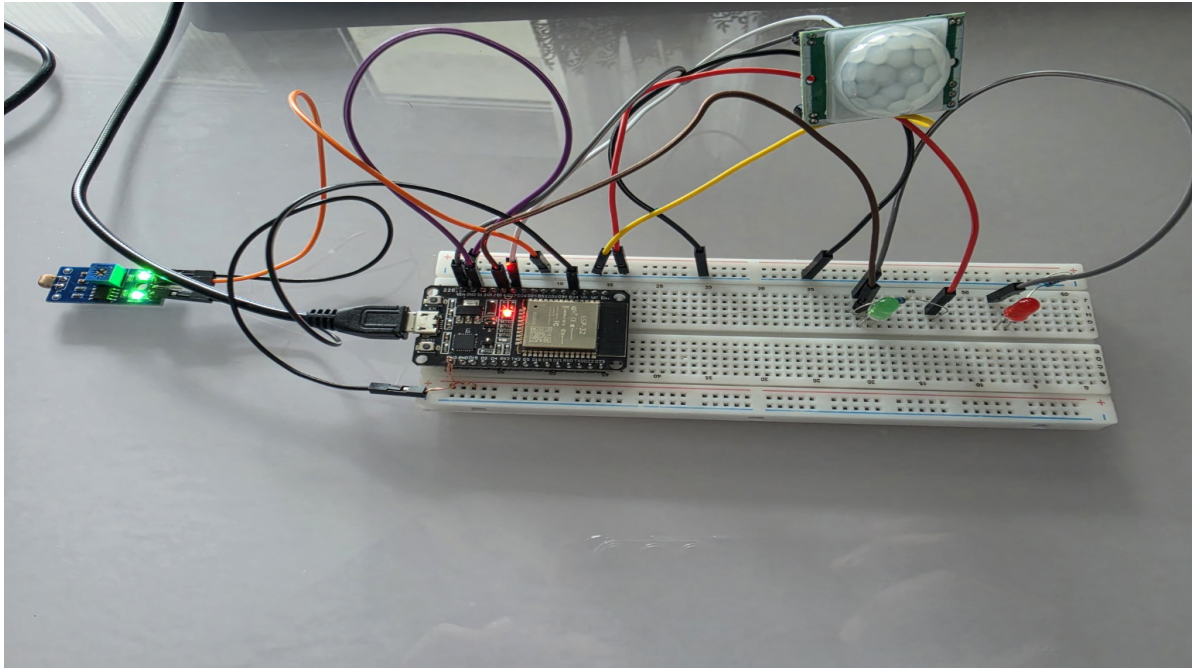
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            else:
                leds.duty(250)
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                print("No Motion -> LEDs DIM")

    print("-----")
    time.sleep(0.5)

```

Project Images

Bright Environment - LEDs OFF



Dark Environment + Motion - LEDs ON / Maximum Brightness

